

SIMATS ENGINEERING

CO5-ASSESSMENT TOOL3-Reverse Engineering

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

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COURSE OUTCOME COVERED

CO5: Design intelligent agents and real-time AI-based systems (chatbots, expert systems, emotion detection, edge AI, autonomous drones, and related applications) by selecting appropriate agent architectures, knowledge representation, and reasoning mechanisms to solve real-world problems. (BL6)

Assessment Tool Weightage: 20%

QUESTIONS: 5

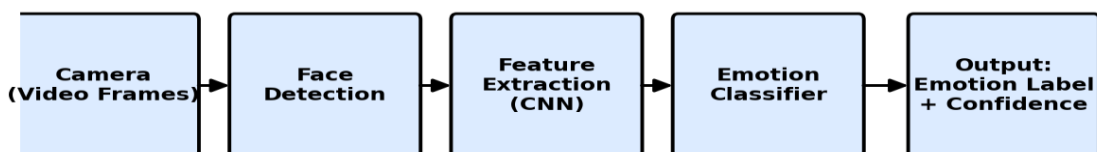
Total Marks:50

Q1. The following is a conversation between a user and an AI-based flight-booking chatbot. Study the transcript given below and reverse-engineer the chatbot's design by: (a) writing its PEAS description, (b) classifying the type of agent it represents with justification, (c) identifying the likely component (NLU, Dialogue Manager, Knowledge Base, or NLG) responsible for each chatbot response, and (d) suggesting one enhancement using a learning-agent architecture.

Speaker	Message
User	I want to book a flight to Chennai next Friday.
Chatbot	Sure! What time would you like to depart, and do you have a preferred airline?
User	Morning flight, no preference.
Chatbot	I found 3 morning flights to Chennai on Friday. Shall I show you the cheapest option first?

Q2. The block diagram below shows the processing pipeline of a Real-Time Emotion Detection System. Reverse-engineer the system by: (a) explaining the function performed at each stage of the pipeline, (b) identifying the likely AI/ML technique used at each stage, (c) writing the PEAS description of the system, and (d) classifying the

Real-Time Emotion Detection System — processing pipeline



agent type (e.g., simple reflex, model-based) with justification.

Q3. An Expert System Shell for medical diagnosis contains the rule base, input facts, and reasoning trace given below. Study it and reverse-engineer the shell by: (a) identifying the type of chaining (forward/backward) used to reach the diagnosis, with justification, (b) identifying the expert-system-shell components illustrated (knowledge base, inference engine, explanation facility), (c) explaining how the shell would perform knowledge acquisition to add a new rule for dengue, and (d) explaining, with reasoning, whether backward chaining could reach the same diagnosis.

Rule	IF (condition)	THEN (conclusion)
R1	fever = yes AND cough = yes	suspect = flu
R2	suspect = flu AND body_ache = yes	diagnosis = Influenza
R3	fever = yes AND rash = yes	suspect = measles

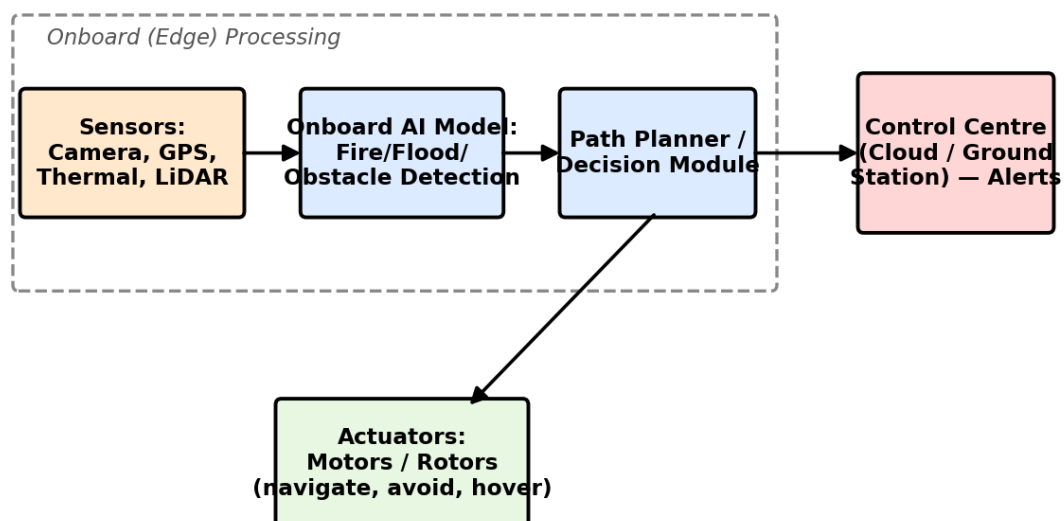
Given facts: fever = yes, cough = yes, body_ache = yes

Reasoning trace: R1 fires first (fever & cough matched) → suspect = flu. R2 then fires (suspect = flu & body_ache matched) → diagnosis = Influenza.

Explanation generated by the shell: “Diagnosis = Influenza because fever and cough indicated flu, and flu together with body ache confirmed Influenza.”

Q4. The diagram below shows the system architecture of an Autonomous Drone used for Disaster Detection with Edge AI deployment. Reverse-engineer the design by: (a) writing the PEAS description of the drone agent, (b) classifying the agent type (goal-based/utility-based) with justification, (c) identifying which components run on the edge versus the cloud and justifying why on-edge processing was chosen for detection and obstacle avoidance, and (d) identifying whether the agent's control architecture is reactive, deliberative, or hybrid, with justification.

Autonomous Drone — Disaster Detection (Edge AI) system architecture



Q5. The table below shows sample output logs from an AI-based Deepfake & Face Authenticity Verification system across five video frames. Study the output and reverse-engineer the system by: (a) identifying the likely input features used by the model (e.g., blink rate, texture/artifact score), (b) identifying the type of model/technique likely used to produce these outputs, (c) explaining how the final Real/Fake decision may have been derived from the frame-level confidence scores,

and (d) suggesting one way a swarm/ensemble-based approach could improve the system's reliability.

Frame No.	Blink Rate (per 10s)	Texture/Artifact Score	Fake Confidence (%)	Prediction
1	4	0.12	8	Real
2	1	0.61	74	Fake
3	0	0.83	91	Fake
4	5	0.09	5	Real
5	2	0.55	68	Fake

Overall system verdict: Fake (3 of 5 frames flagged — majority vote)

Code of Conduct:

I P.Ramesh (192312443) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Ramesh

RUBRICS FOR CALCULATION

Criteria	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning	Max Marks
Identification of Agent Type & PEAS (correctly inferred from the given artifact)	Correctly identifies the agent type and gives a complete, accurate PEAS description, fully consistent with the given artifact.	Identifies the agent type and PEAS description correctly with only minor omissions.	Partially correct identification of agent type/PEAS; some elements are missing or inconsistent with the artifact.	Agent type/PEAS identification is largely incorrect or not attempted.	10
Identification of Architecture & Components (mapping artifact evidence to system components)	Accurately identifies all relevant architectural components and correctly maps each to specific evidence in the artifact.	Identifies most components correctly with reasonable mapping to the artifact; minor gaps present.	Identifies some components; mapping to the artifact is weak or partially incorrect.	Little or no correct identification of architecture/components.	10
Identification of Underlying Technique/Algorithm/Reas	Correctly reverse-engineers the technique, algorithm, or	Correctly identifies the technique/mechanism	Identification is only partially correct or lacks	Technique/mechanism is incorrectly identified or not addressed.	10

Criteria	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning	Max Marks
oning Mechanism	reasoning mechanism (e.g., chaining type, ML technique) with clear supporting evidence.	with adequate, mostly accurate justification.	sufficient supporting evidence.		
Quality of Justification & Use of Evidence (linking conclusions back to the given artifact)	Every conclusion is clearly justified with specific, accurate references to the given artifact.	Most conclusions are justified with reasonable references to the artifact.	Justification is vague or only loosely connected to the artifact.	Conclusions are stated with little or no justification or evidence.	10
Clarity & Completeness of Presentation (addresses all sub-parts of the question in an organised manner)	All sub-parts are addressed completely, in a clear, well-organised, and logically structured manner.	Most sub-parts are addressed clearly with minor gaps in organisation or completeness.	Some sub-parts are missing or the presentation is unclear/poorly organised.	Response is incomplete, disorganised, or largely missing.	10
				Total	50

1)ANS:

Flight-Booking Chatbot :

(a)PEAS Description

Component	Description
P – Performance	Correct flight search, fast response, suitable options, user satisfaction
E – Environment	Users, airlines, airports, flight schedules, prices, booking system
A – Actuators	Chat responses, flight recommendations, booking/confirmation actions
S – Sensors	User messages, flight databases, availability and price information

(b) Type of Agent

The chatbot is mainly a Goal-Based Agent.

Justification:

- The user's goal is to book a flight to Chennai.
- The chatbot asks questions to obtain missing information such as departure time and airline preference.
- It searches available flights and recommends an option.
- Its actions are selected based on achieving the user's booking goal.

It also has some characteristics of a utility-based agent, because it considers options such as the cheapest flight.

(c) Component Responsible for Each Response

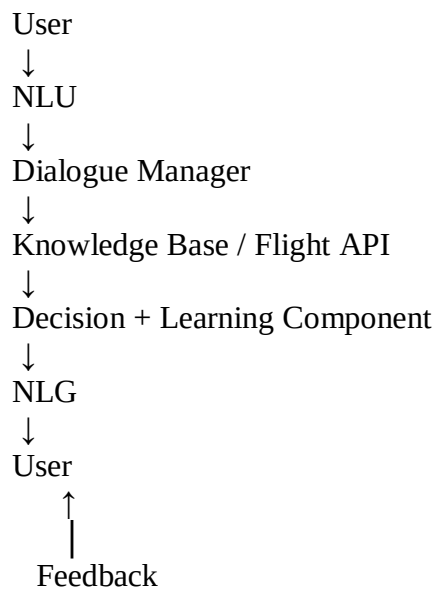
Chatbot Response	Main Component	Reason
“Sure! What time would you like to depart, and do you have a preferred airline?”	Dialogue Manager	Identifies missing booking information and asks follow-up questions.
“I found 3 morning flights to Chennai on Friday.”	Knowledge Base	Uses flight availability/schedule information.
“Shall I show you the cheapest option first?”	Dialogue Manager	Selects the next action based on the user's preference and booking goal.
Natural-language wording of responses	NLG	Converts the system's decisions into understandable language.

NLU role: It interprets user inputs such as “Chennai next Friday”, “Morning flight”, and “no preference.”

(d) Learning-Agent Enhancement

Add a learning component that learns from previous user choices

Proposed Architecture



Enhancement

The chatbot could learn user preferences such as:

- Preferred airlines
- Preferred departure times
- Preference for cheapest flights

Conclusion

The chatbot is a goal-based conversational agent using NLU, Dialogue Management, Knowledge Base, and NLG. Adding a learning component would make it more personalized, adaptive, and efficient.

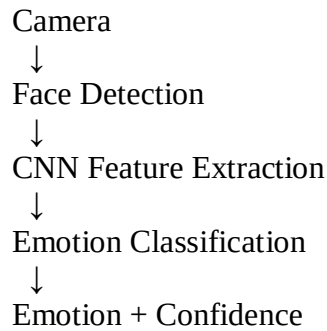
2)ANS:

Real-Time Emotion Detection System :

(a) Function of Each Stage

Stage	Function
Camera	Captures real-time video frames of the person's face.
Face Detection	Detects and extracts the face region from each frame.
Feature Extraction (CNN)	Extracts facial features such as eyes, mouth, and facial expressions.
Emotion Classifier	Classifies the extracted features into emotions such as happy, sad, angry, or neutral.
Output	Displays the predicted emotion label and confidence score.

Processing Flow



(b) Likely AI/ML Techniques

- Camera: Computer vision/video processing.
- Face Detection: Haar Cascade, HOG, or a CNN-based face detector.
- Feature Extraction: Convolutional Neural Network (CNN).
- Emotion Classification: Softmax-based neural network, SVM, or another classifier.
- Confidence: Classification probability generated by the model.

(c) PEAS Description

PEAS	Description
P – Performance	High emotion classification accuracy, low response time, reliable predictions
E – Environment	Real-world surroundings, people, lighting, facial expressions
A – Actuators	Display emotion label, confidence score, or trigger an application response
S – Sensors	Camera/video frames

(d) Agent Type

The system can be classified mainly as a Model-Based Reflex Agent.

Justification

- It receives current visual information from the camera.
- It processes the detected face and extracts features.
- It maintains a representation of the current input while performing recognition.
- It uses the learned CNN/classification model to determine the appropriate emotion.

However, if each frame is classified independently without maintaining any internal state, it is more accurately described as a simple reflex-style perception system.

Agent Type: Model-Based Reflex Agent, because the system uses the current percept (facial image) together with an internal learned model to produce an emotion prediction

3)ANS:

Expert System Shell :

(a) Type of Chaining

The system uses Forward Chaining.

Reason:

- It starts with the given facts:
fever = yes, cough = yes, body_ache = yes.
- R1 matches → suspect = flu.
- The newly derived fact triggers R2 → diagnosis = Influenza.
- Therefore, reasoning proceeds from facts toward the conclusion.

Facts

↓
R1

↓
suspect = flu

↓
R2
↓
Diagnosis = Influenza

(b) Expert-System Shell Components

Component	Example from the system
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Knowledge Base	Rules R1, R2, R3 and medical facts
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Inference Engine	Matches facts with rules and fires R1 and R2
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Explanation Facility Generates: “Diagnosis = Influenza because...”

User Interface	Accepts patient facts and displays diagnosis
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(c) Knowledge Acquisition for Dengue

A medical expert provides the new dengue rule to the knowledge engineer.

Example:

R4:

IF fever = yes AND rash = yes AND body_ache = yes

THEN suspect = dengue

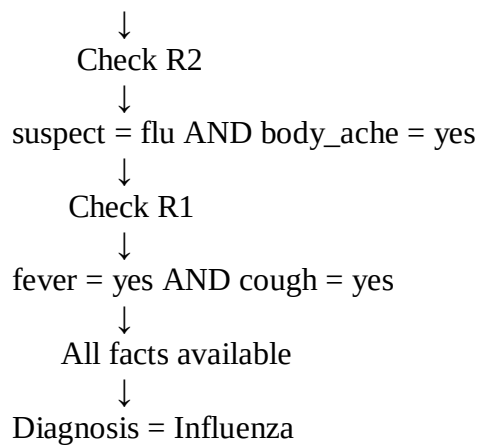
The knowledge-acquisition process is:

1. Consult the medical expert.
2. Identify symptoms associated with dengue.
3. Convert the expert knowledge into an IF–THEN rule.
4. Add R4 to the knowledge base.
5. Test the rule using sample patient facts.
6. Validate the result with the medical expert.

(d) Can Backward Chaining Reach the Same Diagnosis

Backward chaining starts from the goal:

Goal: diagnosis = Influenza



Conclusion

Backward chaining can reach the same diagnosis, but its direction is different:

- Forward chaining: Facts → Rules → Diagnosis
- Backward chaining: Diagnosis → Required rules → Facts

The given reasoning trace specifically demonstrates forward chaining.

4)ANS:

Autonomous Drone Disaster Detection with Edge AI:

(a) PEAS Description

PEAS	Description
P – Performance	Accurate disaster detection, safe navigation, fast response, low energy usage
E – Environment	Disaster areas such as flood, fire, earthquake zones, buildings, obstacles
A – Actuators	Motors and rotors for navigation, avoidance, hovering
S – Sensors	Camera, GPS, thermal sensor, LiDAR

(b) Agent Type: Utility-Based Agent

The drone is mainly a utility-based agent.

Justification:

- It must achieve goals such as detecting disasters and reaching target areas.
- It also needs to choose the best and safest path.
- It balances factors such as safety, speed, energy consumption, and detection accuracy.
- Therefore, it selects actions that maximize overall utility.

Edge vs Cloud Components (c)

Edge / Onboard

- Camera
- GPS

- Thermal sensor
- LiDAR
- Fire/Flood/Obstacle Detection AI model
- Path Planner / Decision Module
- Motor and rotor control

Cloud / Ground Control Centre

- Disaster alerts
- Remote monitoring
- Data storage
- Detailed analysis
- Mission reports

Why Edge AI

- Low latency: Immediate obstacle detection.
- Faster response: No need to wait for cloud processing.
- Works with poor connectivity: Drone can operate even when the network is unavailable.
- Improved safety: Quick decisions prevent collisions.
- Reduced bandwidth: Only important results/alerts need to be transmitted.

Sensors → Edge AI → Path Planner → Actuators
 ↓
 Control Centre
 (Cloud)

(d) Control Architecture

The drone uses a Hybrid Control Architecture.

Reactive Control

- Quickly responds to obstacles detected by LiDAR/camera.
- Example: obstacle detected → immediately change direction.

Deliberative Control

- Path Planner decides the best route toward the disaster location.
- Considers mission objectives and navigation constraints.

Conclusion

Because the system combines fast reactive obstacle avoidance with planned route/mission decisions, it is a hybrid architecture.

5)ANS:

Deepfake & Face Authenticity Verification – Reverse Engineering:

(a) Likely Input Features

- **Blink Rate:** Detects abnormal or unnatural blinking patterns.
- **Texture/Artifact Score:** Detects facial inconsistencies, compression artifacts, or generated-image patterns.
- **Facial landmarks:** Eye, nose, and mouth positions.
- **Skin/face texture:** Identifies unnatural details.
- **Temporal consistency:** Checks whether facial features change naturally between frames.

(b) Likely Model/Technique

The system likely uses a CNN-based deepfake classifier.

- CNN extracts facial visual features.
- A classifier produces Fake Confidence (%).
- Video frames can be analyzed individually and then combined.
- A threshold such as 50% can be used:
 - Fake confidence $\geq 50\% \rightarrow$ Fake
 - Fake confidence $< 50\% \rightarrow$ Real

(c) Final Real/Fake Decision

The frame-level results are:

Frame Fake Confidence Prediction

1	8%	Real
2	74%	Fake
3	91%	Fake
4	5%	Real
5	68%	Fake

Average fake confidence:

Although the average is just below 50%, 3 out of 5 frames are classified as Fake.

Therefore, a likely video-level decision rule is majority voting, giving:

Final Prediction = FAKE

(d) Swarm/Ensemble-Based Improvement

Use an ensemble of multiple independent deepfake detectors.

Video



Voting / Fusion



Final Real/Fake

- Different models analyze different features.
- Their predictions are combined using majority voting or weighted averaging.
- This reduces the effect of an incorrect prediction from a single model.
- It improves robustness and reliability of deepfake detection.

Conclusion: The system likely combines facial/temporal features + CNN classification + frame-level aggregation to make the final authenticity decision.